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CHEMISTRY
9202/2
Paper 2

Mark scheme

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Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from aqa.org.uk

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

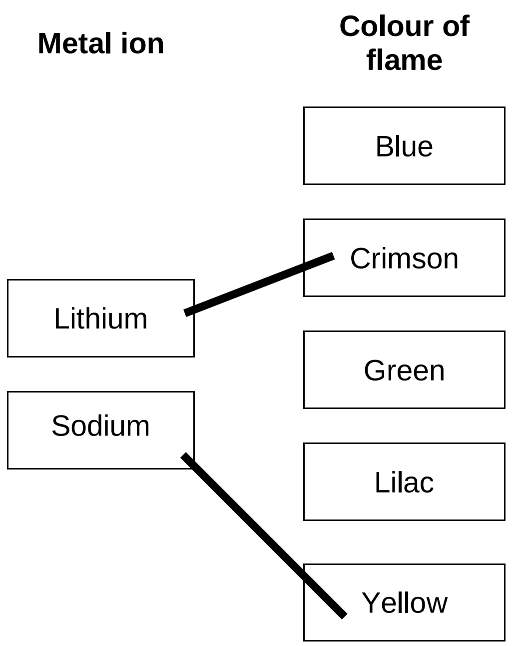
Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Answers	Extra information	Mark	AO/Spec. Ref.	ID
01.1	3		1	AO2 3.1.2.e/f	A
01.2	dip wire into solution hold wire in blue flame (of Bunsen burner)	ignore comments about cleaning wire do not accept hold wire in yellow flame (of Bunsen burner) ignore observe colour of flame	1 1	AO1 3.4.3.a	E
01.3			1 1	AO1 3.4.3.a	E
01.4	(as you go down the group) melting point decreases (as you go down the group) density increases except for potassium / sodium	ignore atomic number	1 1	AO3 3.7.1.a	E
01.5		an answer of 0.5 (cm^3) scores **2** marks		AO2 3.7.1.a	E

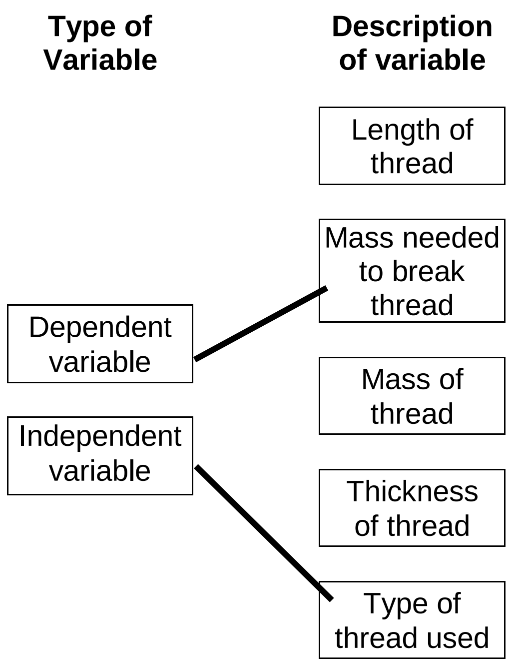
	(volume =) $\frac{0.43}{0.86}$ = 0.5 (cm^3)		1 1		

01.6	sodium + water → sodium hydroxide + hydrogen	allow $2Na + 2H_2O \rightarrow 2NaOH + H_2$ allow **1** mark for hydrogen / H_2 (as a product)	2	AO1/AO2 3.6.1.a 3.7.1.a	E

01.7	14		1	AO1 3.5.1.f	A

01.8	any **one** from: • wear gloves • wear goggles		1	AO4 3.5.1.a	E

Total			**13**

Question	Answers	Extra information	Mark	AO/Spec. Ref.	ID
02.1	 (dependent variable) mass needed to break thread (independent variable) type of thread used.		1 1	AO4 3.10.2.d	E
02.2	nylon attempt 3	both required for **1** mark	1	AO2 3.10.2.d	E
02.3	any **one** from: • thread was thinner • thread was damaged	do **not** accept thread was smaller	1	AO3 3.10.2.d	E
02.4	any **one** from: • use smaller masses • repeat **and** recalculate the mean • repeat **and** remove anomalous results		1	AO4 3.10.2.d	E
02.5	any **one** from: • lack of space in landfill • litter • harm to wildlife	ignore pollution	1	AO1 3.10.2.e	E

02.6	(there are) weak (intermolecular) forces between the (polymer) chains (so) little energy is required to overcome the weak / intermolecular forces	allow (there are) no cross-links between the (polymer) chains	1 1	AO4 3.10.2.c	E

Total			**8**

Question	Answers	Extra information	Mark	AO/Spec. Ref.	ID
03.1	**test:** (add) barium chloride solution (and dilute hydrochloric acid) **result:** white precipitate	allow (add) barium nitrate solution (and dilute nitric acid)	1 1	AO1 3.4.3.f	E
03.2	(white anhydrous) copper sulfate turns blue		1	AO2 3.9.1.d	E
03.3	copper does not react with sulfuric acid		1	AO2 3.5.2.a	E
03.4	react with all the acid	allow neutralise all of the acid	1	AO4 3.5.2.a	E
03.5	to remove excess copper oxide		1	AO4 3.5.2.a	E
03.6	copper sulfate is less soluble at lower temperatures (so) not all the copper sulfate will dissolve in the water		1 1	AO3 3.5.2.b	E

03.7	(solubility at 40 °C =) 29 (g/100cm^3)		1	AO2 AO3 3.5.2.b 3.9.1.d	E
	$\frac{16}{29} \times 100$	allow correct use of an incorrect value for solubility at 40 °C	1		
	= 55 (cm^3)	allow 55.1724137 (cm^3) correctly rounded to at least 2 significant figures	1		

Total			**11**

Question	Answers	Extra information	Mark	AO/Spec. Ref.	ID
04.1	any **three** from: • volume of silver nitrate solution • concentration of silver nitrate solution • mass of metal • surface area of metal	ignore temperature	3	AO4 3.3.1.1.b	E
04.2	(highest temperature) 20.5 (°C) (starting temperature = 20.5 – 4.5 =) 16 (°C)	 allow correct use of an incorrect value for the highest temperature	1 1	AO2 3.3.1.1.b	E
04.3	name of metal is a categoric variable		1	AO3 3.3.1.1.b	A
04.4	 less reactive than iron more reactive than copper	incorrect reactivity of gold, magnesium or zinc = **max 1** allow below iron in reactivity series allow above copper in reactivity series if no other mark awarded, allow **1** mark for between iron and copper in reactivity (series)	 1 1	AO3 3.3.1.1.b	E

04.5	(because) gold is less reactive than silver **or** (because) gold cannot displace silver	allow converse statements	1	AO2 3.3.1.1.b 3.3.1.1.c	E

04.6	any **one** from: • would not be safe • too reactive		1	AO2 3.3.1.1.a	E

04.7	Fe^{2+} **and** Ag **2**(Ag^{+}) **and** **2**(Ag)	ignore state symbols	1 1	AO2 3.3.1.1.b	E

Total			**12**

Question	Answers	Extra information	Mark	AO/Spec. Ref.	ID
05.1	petrol is a mixture (of hydrocarbons)	allow petrol contains different substances	1	AO2 3.4.1.a	E

Question	Answers	Mark	AO/Spec. Ref.	ID
05.2	**Level 3:** A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.	5–6	AO1/AO3 3.10.1.2.g 3.10.1.2.d	E
	Level 2: Some logically linked reasons are given. There may also be a simple judgement.	3–4		
	Level 1: Relevant points are made. They are not logically linked.	1–2		
	No relevant content	0		
	Indicative content Reasons for ethanol: • carbon dioxide produced when burnt is absorbed by the sugar cane through photosynthesis • as less net carbon dioxide produced there is less climate change • renewable source for fuel so unlikely to run out • greater volatility / flammability Reasons against ethanol: • land that could be used for food is used for fuel – causing rising food costs • ethanol produced in fermentation is not pure and has to be distilled • destruction of crops or habitats as forests are cleared to grow biofuel crop • takes a long time for plants to grow and get fuel due to batch process Reasons for petrol • uses less fuel per kilometre so the car travels further before refuelling • fuel obtained very quickly through continuous process Reasons against petrol • carbon dioxide produced leads to climate change • more sulfur dioxide produced leads to acid rain • non-renewable fuel so will run out			

05.3	(overall) energy change labelled		1	AO2 3.9.2.d	E

05.4		an answer of 6048 (J) scores **3** marks		AO2 3.9.2.a	E
	(32.5 – 18.1 =) 14.4 (ºC)		1		
	14.4 × 4.2 × 100	allow correct use of incorrectly calculated temperature change	1		
	= 6048 (J)		1		

05.5	0.1 °C		1	AO3 3.9.2.a	E

05.6		an answer of 19 000 (J/g) scores **2** marks		AO2 3.9.2.b	E
	(75.15 – 74.85 =) 0.3 (g)		1		
	($\frac{5700}{0.3}$ =) 19 000 (J/g)	allow correct use of incorrectly calculated mass change	1		

05.7	• **improvement:** stir water		1	AO3 3.9.2.b	E
	• **reason:** evenly distributes heat energy		1		
	or				
	• **improvement:** use of lagging / lid (1)	do **not** accept plastic / polystyrene cup			
	• **reason:** minimises energy loss (1)				

Total			**16**

Question	Answers	Extra information	Mark	AO/Spec. Ref.	ID
06.1	18 (%)		1	AO2 3.3.2.i	E
06.2		reference to incorrect particles **or** incorrect bonding = **max 3**		AO1/2 3.2.1.b 3.2.1.c	E
	aluminium (atom) loses **and** oxygen (atom) gains (electrons)		1		
	3 electrons transferred per aluminium (atom)	allow two aluminium atoms transfer 6 electrons	1		
	2 electrons transferred per oxygen (atom)	allow 6 oxygen electrons transferred to three oxygen atoms	1		
	to gain a full outer shell **or** to gain noble gas structures	allow aluminium ions and oxide ions (produced) are attracted to each other by strong electrostatic forces of attraction	1		
06.3		an answer of 53 (%) scores **3** marks		AO2 3.6.2.b	E
	$\frac{54}{102} \times 100$		1		
	= 52.9411 (%)		1		
	= 53 (%)	allow an answer correctly rounded to 2 significant figures from an incorrect calculation of the percentage by mass	1		
06.4	ions are free to move	allow free moving ions	1	AO1 3.3.2.a	E

06.5	lowers the melting point (of mixture)	do **not** accept acts as catalyst do **not** accept lowers activation energy	1	AO1 3.3.2.i	E

06.6	(aluminium) ions are attracted to the negative electrode / cathode		1	AO1 3.3.2.i 3.3.2.c	E

06.7	2 O^{2-} $4e^{-}$		1 1	AO2 3.3.2.f 3.3.2.i	E

06.8	(because) the electrode is made of carbon which reacts with oxygen **alternative approach** (because) electrode reacts with oxygen (1) to produce carbon dioxide (1)	allow electrode is made of graphite ignore oxidised	1 1	AO1/2 3.3.2.i	E

Total			**15**

Question	Answers	Extra information	Mark	AO / Spec. Ref.	ID
07.1	(conical) flask		1	AO4 3.6.4.b	E
07.2	volume (of hydrogen / gas) not measured correctly	allow incorrect reading of the scale (on the measuring cylinder)	1	AO4 3.8.1.a	E
07.3	contains air		1	AO3 3.8.1.a	E
07.4	points plotted to within 1 mm line of best fit	allow a tolerance of ± half a small square award 1 mark if 3 points correctly plotted	2 1	AO2 3.8.1.a	E
07.5	smaller amount of product formed per unit time	allow gradient of the line decreases	1	AO3 3.8.1.a	E
07.6	concentration of acid decreases **or** fewer particles of acid per unit volume surface area of magnesium decreases	 allow mass of magnesium decreases	1 1	AO2 3.8.1.a 3.8.1.e	E
07.7	no effect (as) number of mols of acid remains the same **or** (as) number of mols of limiting reactant remains the same	allow volume of hydrogen / gas stays the same	1 1	AO3 3.8.1.a	E

07.8		an answer of 0.0023 (moles) scores **4** marks an incorrect answer in one step does not prevent the allocation of marks in subsequent steps		AO2 3.6.4.a 3.6.1.d	E
	(number of moles of hydrochloric acid =) $\frac{17}{1000} \times 0.20$		1		
	= 0.0034		1		
	(number of moles of magnesium that react $= \frac{0.0034}{2}$ =) 0.0017		1		
	(number of moles in excess = 0.0040 – 0.0017 =) 0.0023		1		

Total			**15**